

1 • Shared evidence: garden brief

The border brief: an ecosystem evidence web

Name _____ Date _____

Fictional garden model. Plant counts below are constructed data from one 1 m² sample per patch.

A • Environment: soil moisture and light can affect plants. Aphids feed on plants; ladybirds feed on aphids. B • Selected food links: leaves → caterpillars; leaves → aphids; aphids → ladybirds; caterpillars → blue tits; aphids → blue tits. Arrows run from food to consumer.

Plant species	Patch A count	Patch B count
Daisy	16	5
Clover	4	5
Dandelion	0	5
Plantain	0	5

W1. Name an abiotic factor and explain one interaction with a living organism.

W2. Fewer aphids: predict one possible ladybird effect and say why it is uncertain.

W3. Compare abundance and richness. Can these samples prove the cause?

2 • Supported route: a clear garden message

Use page 1. Word bank: non-living / consumer / two / four / twenty.

S1. Which is an abiotic factor?

- A. Soil moisture
- B. Ladybird feeding

S2. Complete: An ecosystem includes living organisms interacting with each other and the ____ environment.

S3. Draw leaves → aphids → ladybirds. Explain what one arrow means.

S4. Complete: A has ____ species and B has ____ . Each sample contains ____ individual plants.

S5. Give one reason to take more samples before making a site-wide claim.

3 • Standard route: a garden brief

Use the evidence sheet. Each answer should help the team make a careful decision.

N1. Describe one abiotic interaction and one biotic interaction in the garden.

N2. Draw two connected food chains from the model. Predict a possible effect of fewer aphids and explain one uncertainty.

N3. Compare abundance and species richness using values for both samples.

N4. Recommend a next sampling step and explain why the current data cannot prove a maintenance cause.

4 • Stretch route: challenge the headline

Claim: "Patch B's mowing method has doubled the biodiversity of the whole garden."
The evidence is only the one sample per patch on page 1.

T1. Identify two different overclaims and rewrite the headline so that it matches the evidence.

T2. Propose a comparison using repeated equal-area plant samples. Explain how you would select locations and keep effort comparable.

T3. The team wants to remove all aphids. Use the web to explain two possible effects and one limitation of the model.

5 • Exit and reflection

Work independently. A labelled diagram or scribed answer is welcome.

E1. What must an ecosystem description include beyond the community?

E2. What does the arrow aphid → ladybird show?

E3. Why can two samples with equal total abundance have different species richness?

E4. Name one improvement to the sampling and explain its purpose.

Reflection. Which question should the garden team investigate next, and why?
